UNIVERSITY OF LONDON
BSc ALTERNATIVE ASSESSMENT 2021

For Internal Students of

Royal Holloway

IY3609: Digital Forensics

IY3609R: Digital Forensics (PAPER FOR FIRST SITS/RESIT CANDIDATES)

Time Allowed: **TWO hours**

Attempt **three** questions

No credit will be given for attempting any further questions

- Answers must be hand-written with module and candidate numbers at the top of each page. Photograph or scan all pages, and retain paper originals for reference by examiners.
- For each attempted question, clearly state the question number.
- Do not include Student ID Number or name, only candidate number
- **Academic Misconduct:** All assessments will be checked for academic misconduct. Any suspected offences will be dealt with under the College's Academic Misconduct procedures.
  - The work submitted is expected to be your own work and only your work. You may not ask for help with the assessments from any source, or copy anyone else's work.
  - You must not give help to anyone else, including sending them any parts of the questions or copies of your solutions
  - You must not discuss the questions or solutions with anyone else
- **Submitting your work**:
  - Your document must be submitted through Moodle as a single PDF file
  - Emailed submissions will not be accepted
  - The exam must be uploaded within one hour of finishing the exam

© Royal Holloway University of London 2021

Important Copyright Notice
This exam paper has been made available in electronic form
strictly for the educational benefit of current Royal Holloway students
on the course of study in question.
No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

Question 1 Network and Email Forensics

(a) An organised crime syndicate wishes to develop a novel command and control network for their ransomware and DDoS botnet.

An ethically flexible digital forensics expert has been employed to advise on optimising the design to allow evasion of network forensics techniques.

(i) The command and control (C2) infrastructure protocols should allow real-time co-ordination of activities by bots, yet be inconspicuous to a forensic analyst examining network traffic.

Suggest one network or application protocol that could be the used by the C2 architecture as the basis for a steganographic layer that is difficult to detect, and will allow the C2 network to satisfy the above requirements. Give a reasoned answer; it is not necessary to describe protocols or PDUs in detail. [15]

(ii) The ethically flexible expert subsequently advises a forensics laboratory on detection mechanisms for obfuscated network command and control channels.

Which mechanisms for detecting obfuscated (steganographic) channels can be deployed if monitoring is limited to observing network traffic passively, i.e. without interventions at suspect endpoints? Give a reasoned answer. [10]

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

Question 2 Legal Matters

(a) The U.S. Supreme Court *Daubert* cases established a standard for the quality of expert testimony that is also relevant to U.K. courts.

In light of these rulings, consider a criminal case where a digital forensics expert wishes to introduce a novel method for the analysis of a cloud services constellation.

Which questions and challenges is the expert witness likely to confront when presenting the novel techniques to justify a finding, and how could these be addressed? [15]

(b) The UK Overseas Production Order (OPO) establishes the right under UK law to demand electronic evidence from a network service provider based outside the UK for the purposes of criminal investigations.

Consider the case of a Belgian Internet service provider being subjected to an OPO related to a UK national using the Belgian provider. The UK national is accused of committing VAT fraud under section 72(1) of the VAT Act.

What response can the UK court expect from the Belgian service provider? Give a reasoned answer. [10]

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

Question 3 Storage Forensics I

(a) Consider the case of a SSD connected via a SATA interface. The SSD was used extensively over the period of several years. Prior to the decommissioning of the SSD, files are to be recovered to a new device.

However, on copying the files, the SSD shows read errors for many of these files. A forensics expert is hence brought in, and is asked to recover files.

Formulate hypotheses for the forensic analyst to pursue, and provide reasoning for these. Describe how the analyst may seek to recover the information even if the sectors containing the files cannot be accessed by way of the (S)ATA interface. [15]

(b) A hybrid hard drive was recovered at a crime scene, but has suffered extensive physical damage by being submerged for a prolonged period in a lake.

Investigators wish to recover information from the drive; which approach can a forensic analyst take to recover information from the drive? Give a reasoned answer. [10]

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

Question 4 Storage Forensics II

(a) Self-encrypting drives compatible with the TCG Opal specification also support a feature called **Cryptographic Erasure**.

If an adversary (e.g. a law enforcement or intelligence agency) has the ability to place an *implant* on the computer system of a suspect, how can the contents of the drive be preserved for investigation even when the owner of the drive issues the command to erase cryptographically? Give a reasoned answer outlining assumptions and a scenario. [15]

(b) Consider a situation where a law enforcement agency wishes to place an implant on a target system to obtain memory images (e.g. for obtaining both plaintext data and authentication keys as well as encryption keys for cloud services) from a target system.

However, the target has activated the **Secure Boot** facility on on the objective.

Can the law enforcement specialist still place an implant in the target, particularly when given only a few minutes of unobserved physical interaction with the target system? Give a reasoned answer and provide a detailed scenario. [10]

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.

Question 5 Memory Forensics

(a) A forensic investigator suspects that a laptop computer running Microsoft Windows 10 is compromised and has had a root kit and remote administration toolkit (RAT) installed. The system is running and connected to the network.

(i) Identify one technical means fo the investigator to take a memory image of the machine, assuming that there is no memory snapshot agent installed already.

Give a step-by-step account of how the image is to be taken, what level of fidelity can be achieved by your chosen technique, and detail any limitations that the technique may have. [13]

(ii) For the technique described in the answer to question 5(a)(i), describe whether or not and how the rootkit can detect and counteract the taking of a memory image.

Give a reasoned answer on how effective the countermeasures will be. [12]

END

SW

No further copying, distribution or publication of this exam paper is permitted.
By printing or downloading this exam paper, you are consenting to these restrictions.